

Data Center RDIMM DDR5 Tester

HyperRAM



File: hyperram.kicad_sch

DDR5



File: DDR5.kicad_sch

Interfaces



File: interfaces.kicad_sch

Ethernet



File: ethernet.kicad_sch

Supply



File: supply.kicad_sch

Config SPI flash



File: config-spi.kicad_sch

FPGA power



File: fpga-power.kicad_sch

FPGA banks 12-15



File: fpga-banks-12-15.kicad_sch

FPGA banks 16-34



File: fpga-banks-16-34.kicad_sch



www.antmicro.com

Antmicro

Sheet: /
File: data-center-rdim--ddr5-tester.kicad_sch

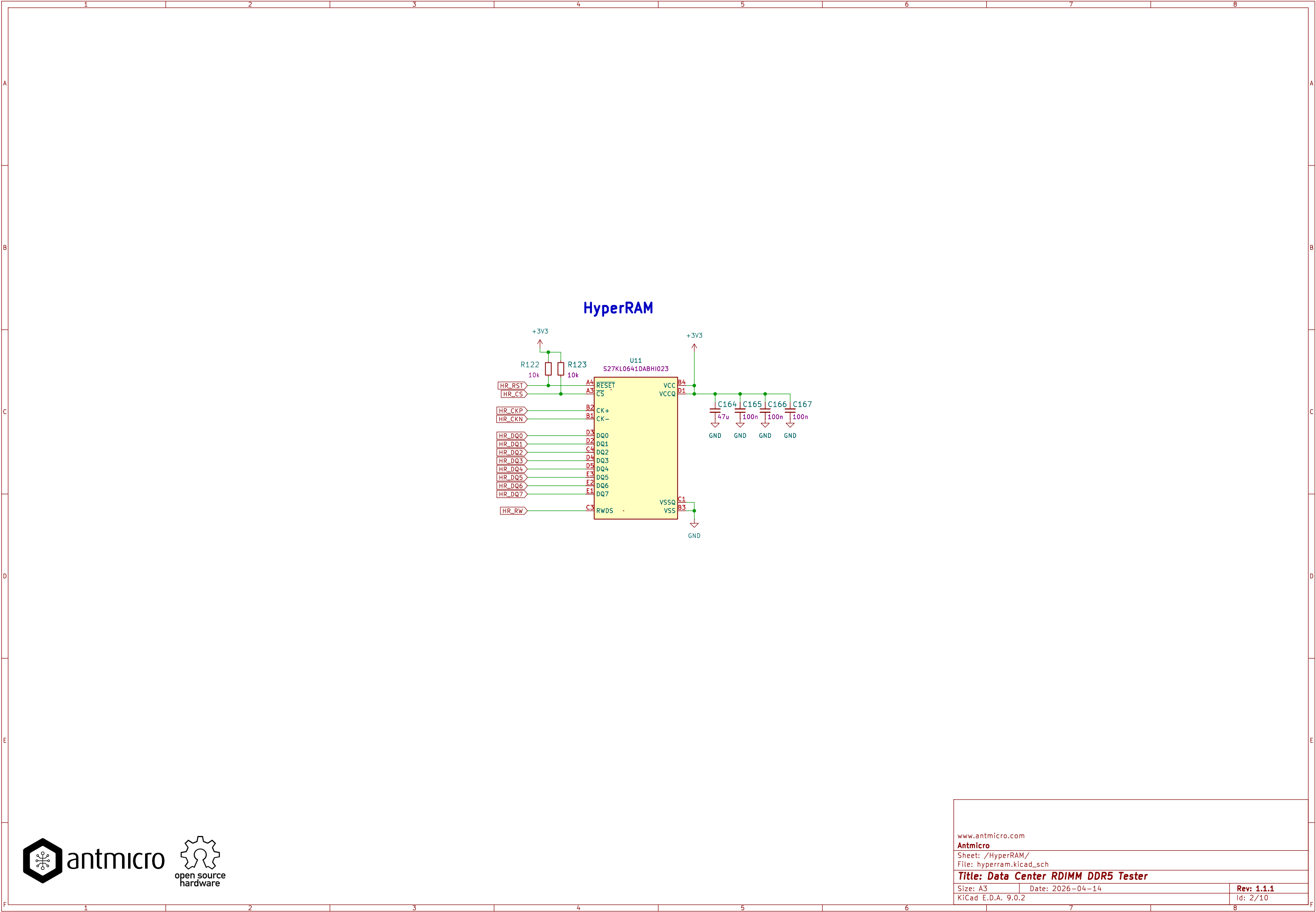
Title: Data Center RDIMM DDR5 Tester

Size: A3
KiCad E.D.A. 9.0.2

Date: 2026-04-14

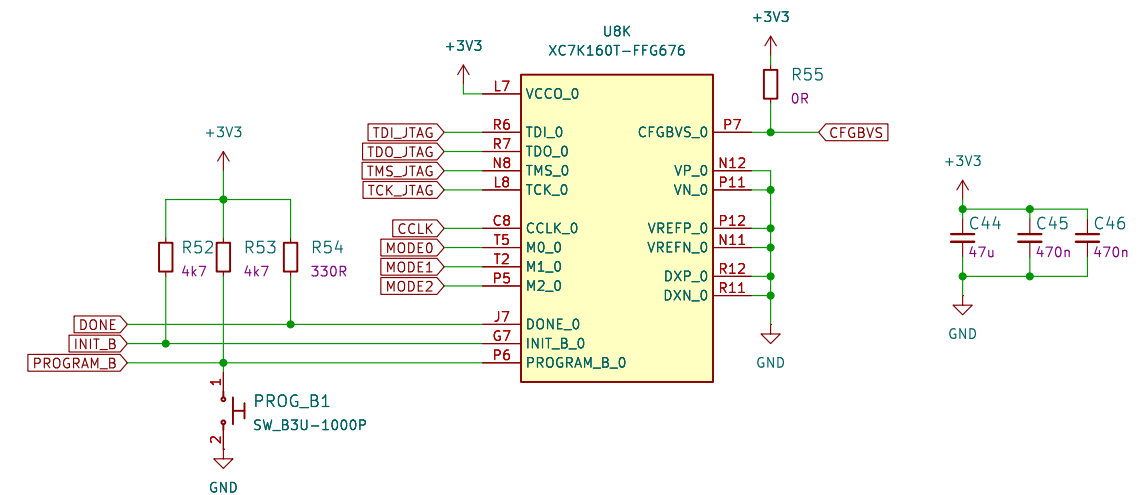
Rev: 1.1.1

Id: 1/10

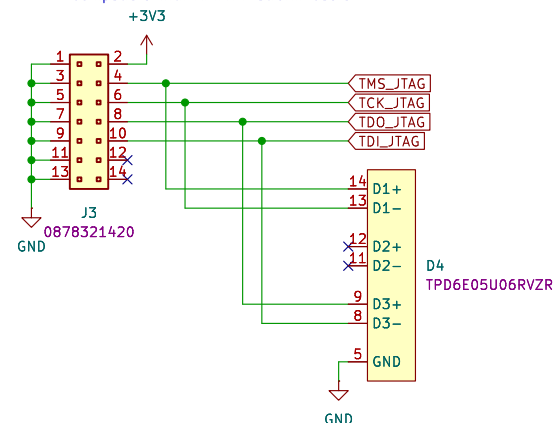


Follows Figure 2–14 7 Series FPGAs Configuration User Guide
UG470 (v1.13.1)

TODO: verify after FPGA swap

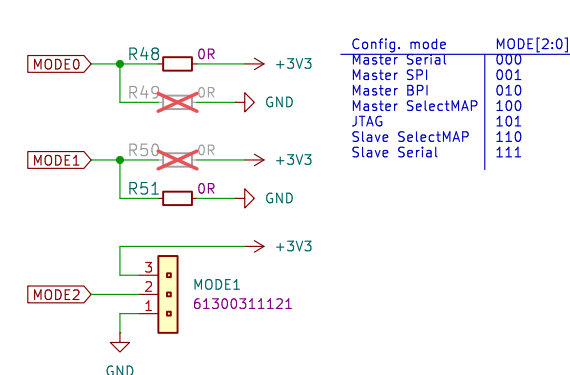


Compatible with Xilinx Platform Cable

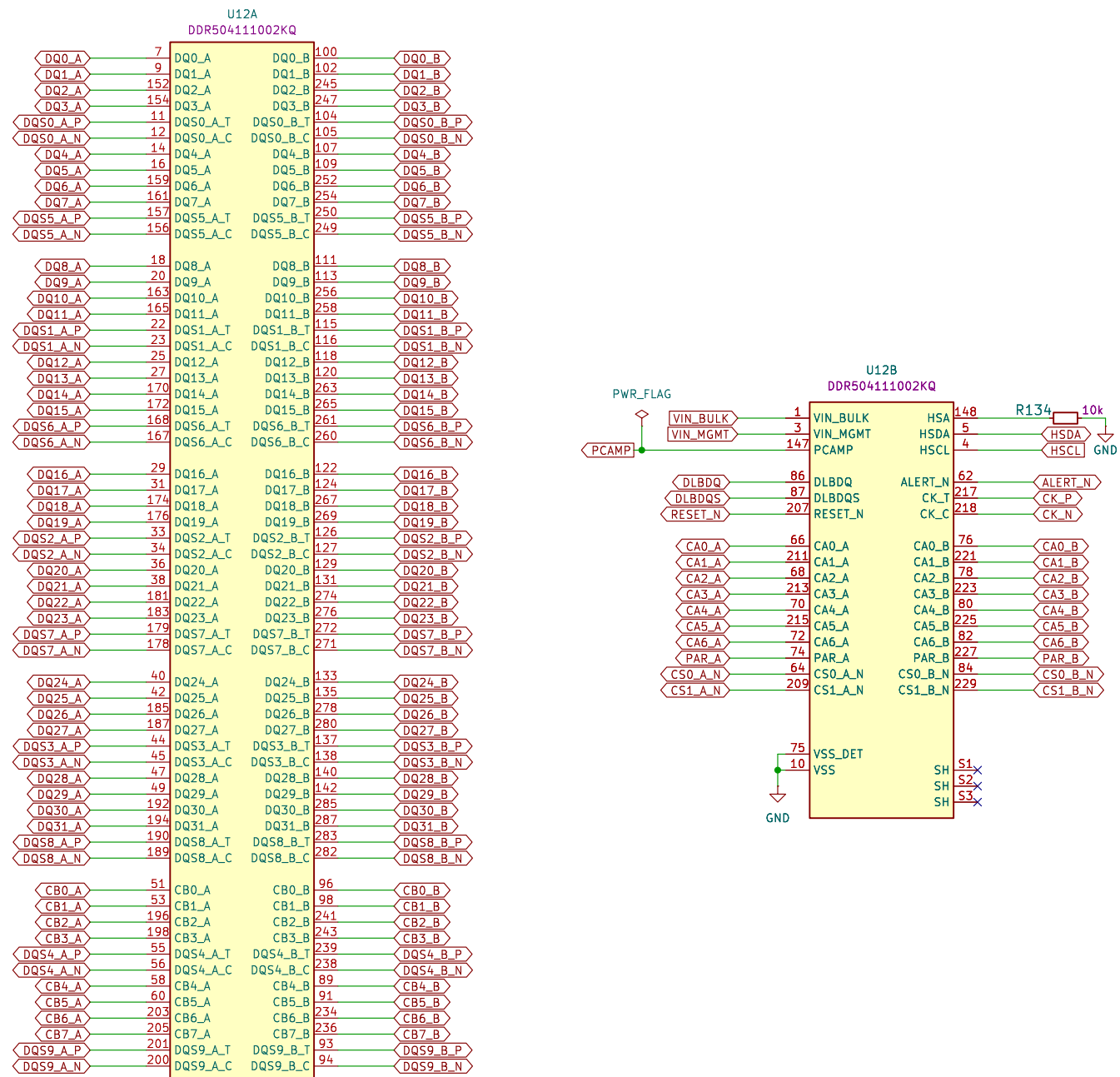


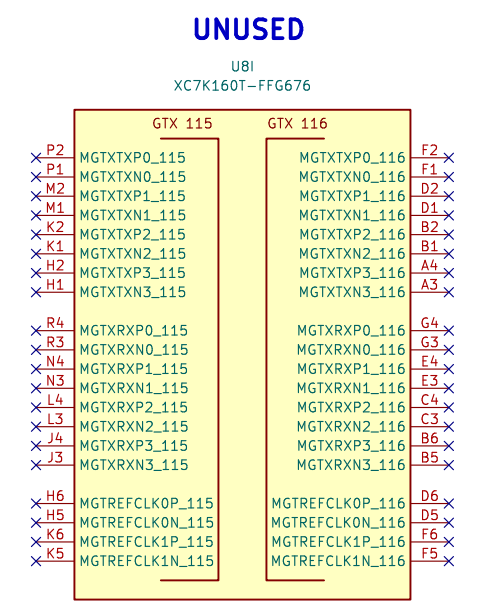
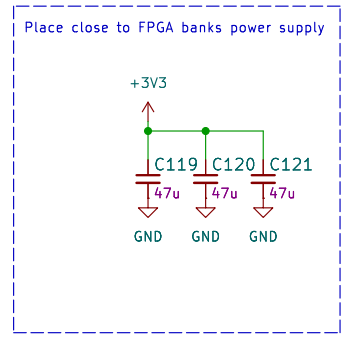
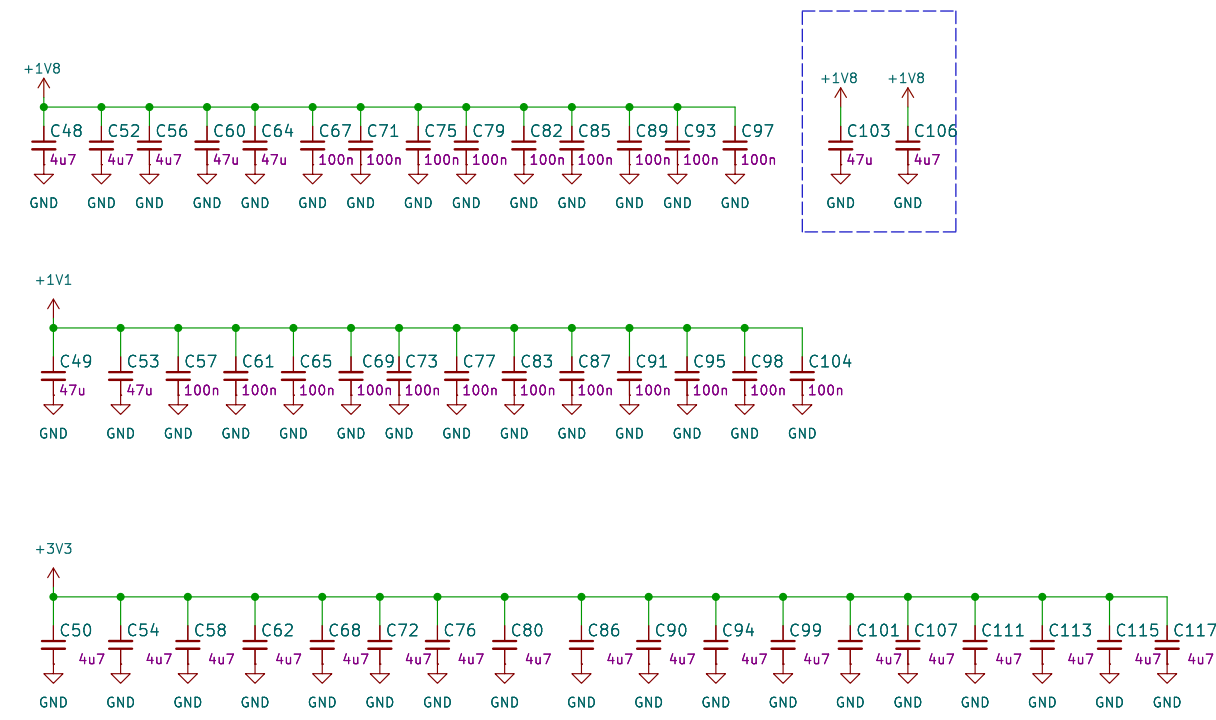
The image shows two identical circuit diagrams for LEDs D5 and D6. Each circuit consists of a green LED (D5 or D6) connected in series with a 330R resistor (R46 or R47) to a +3V3 supply. The LEDs are controlled by signals DONE and INIT_B, respectively. The LEDs are connected to the D pin of a BSS138PW MOSFET (Q1 or Q2), which is also connected to GND. The MOSFETs are controlled by signals DONE and INIT_B, respectively. The MOSFETs are connected to the S pin of a BSS138PW MOSFET (Q1 or Q2), which is also connected to GND. The MOSFETs are connected to the S pin of a BSS138PW MOSFET (Q1 or Q2), which is also connected to GND.

For details, see UG470 p. 21

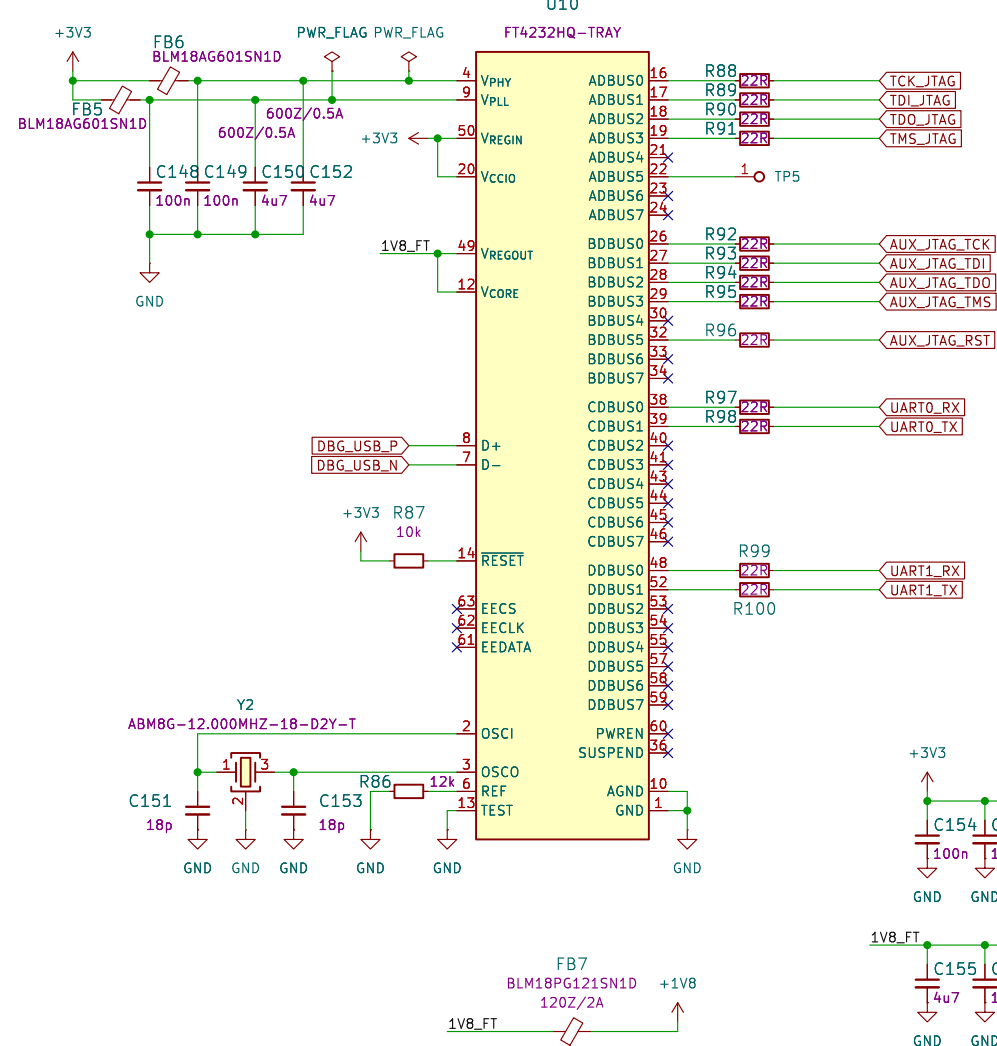


DDR5 RDIMM connector

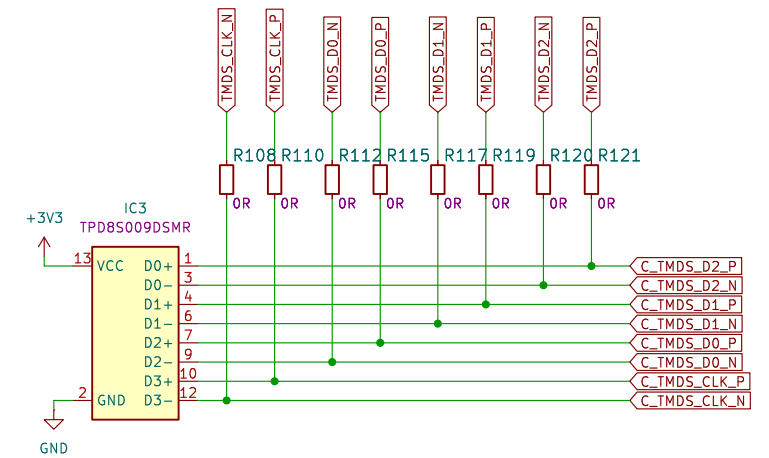
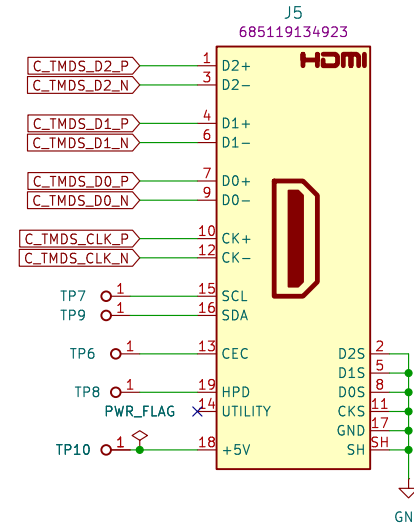




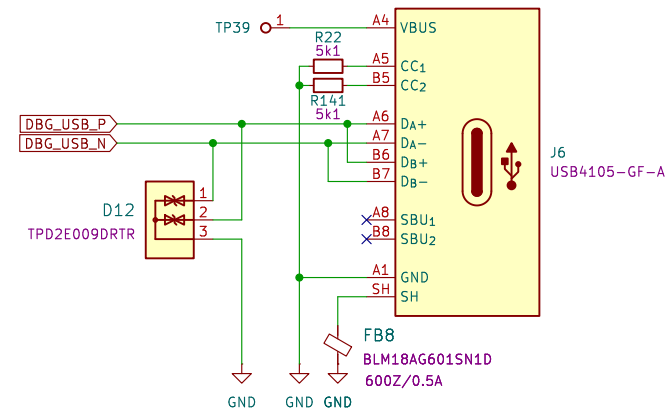
Debug UART



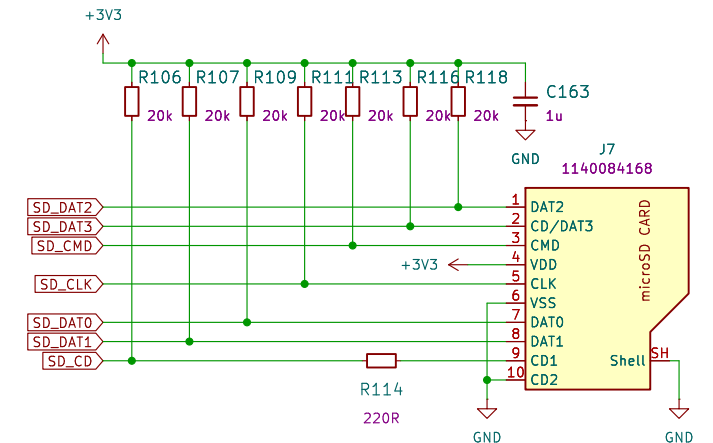
HDMI



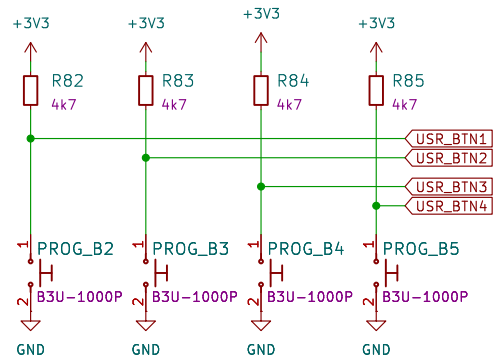
Debug USB



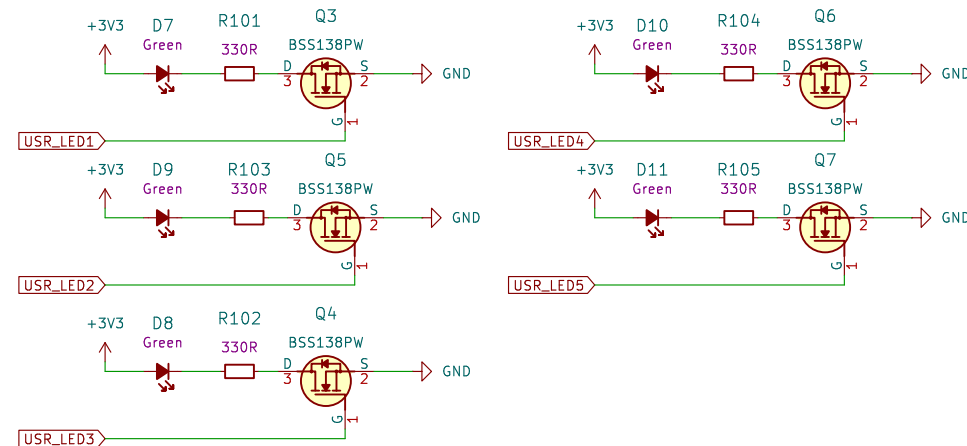
SD card



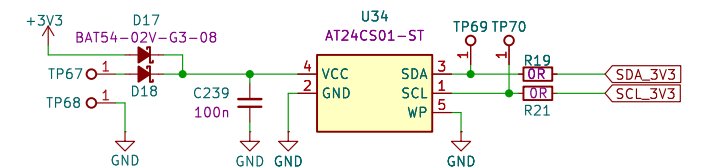
User buttons

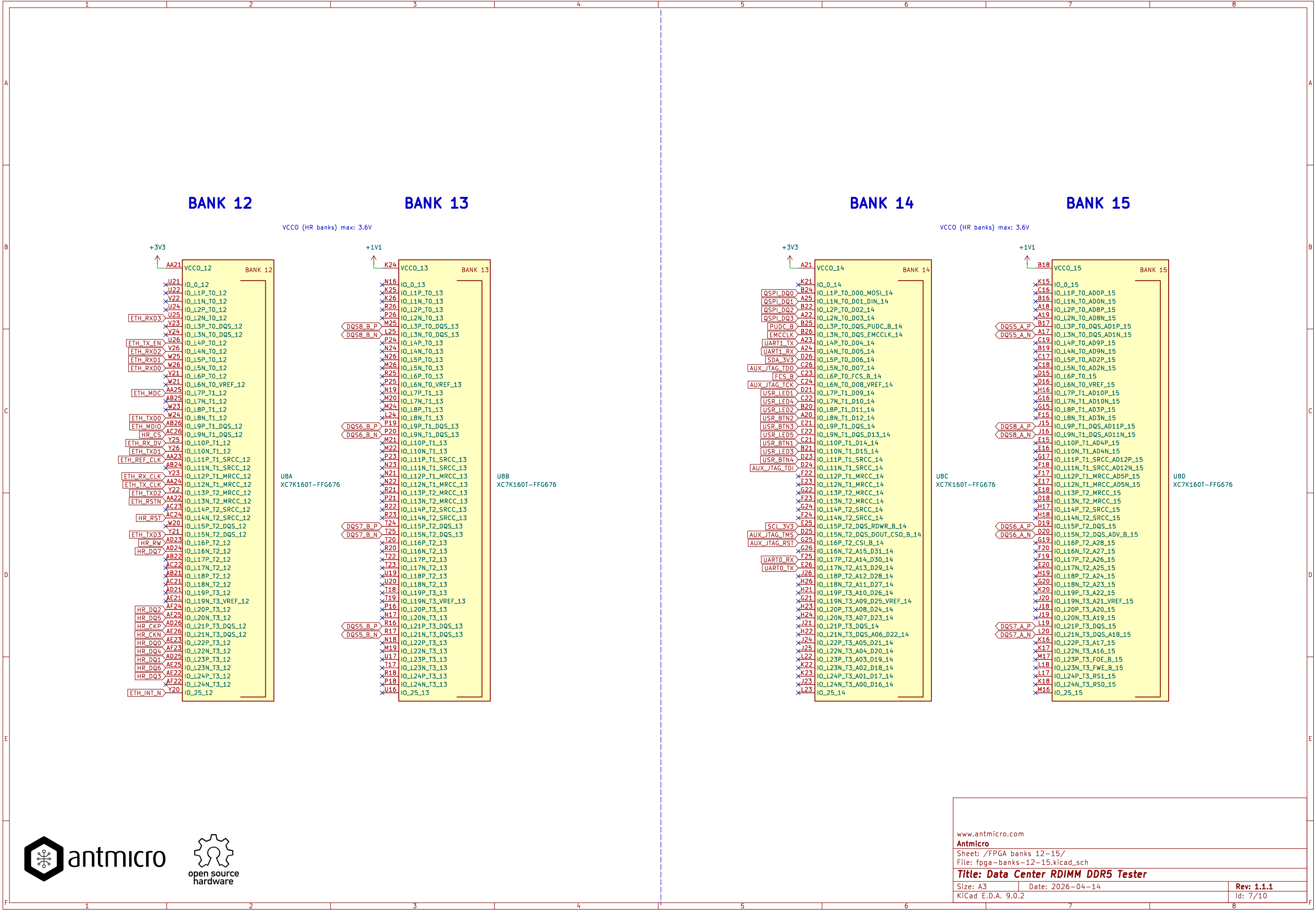


User LEDs

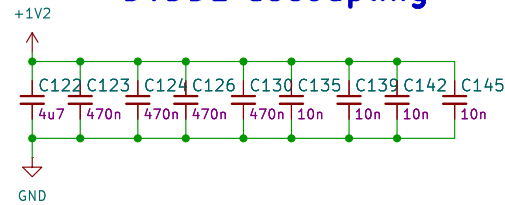


UID EEPROM

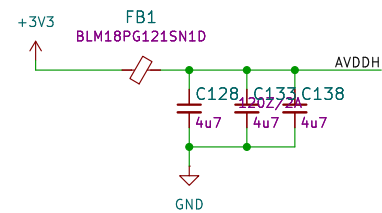




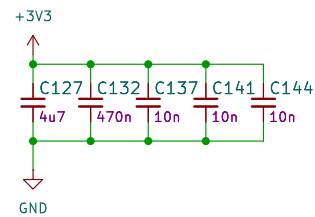
DVDDL decoupling



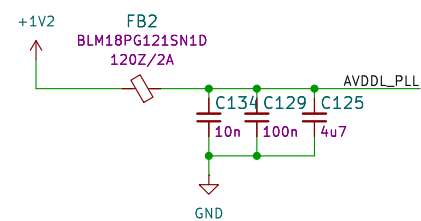
AVDDH decoupling



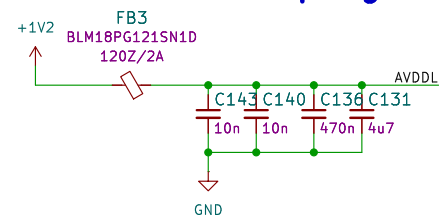
DVDDH decoupling



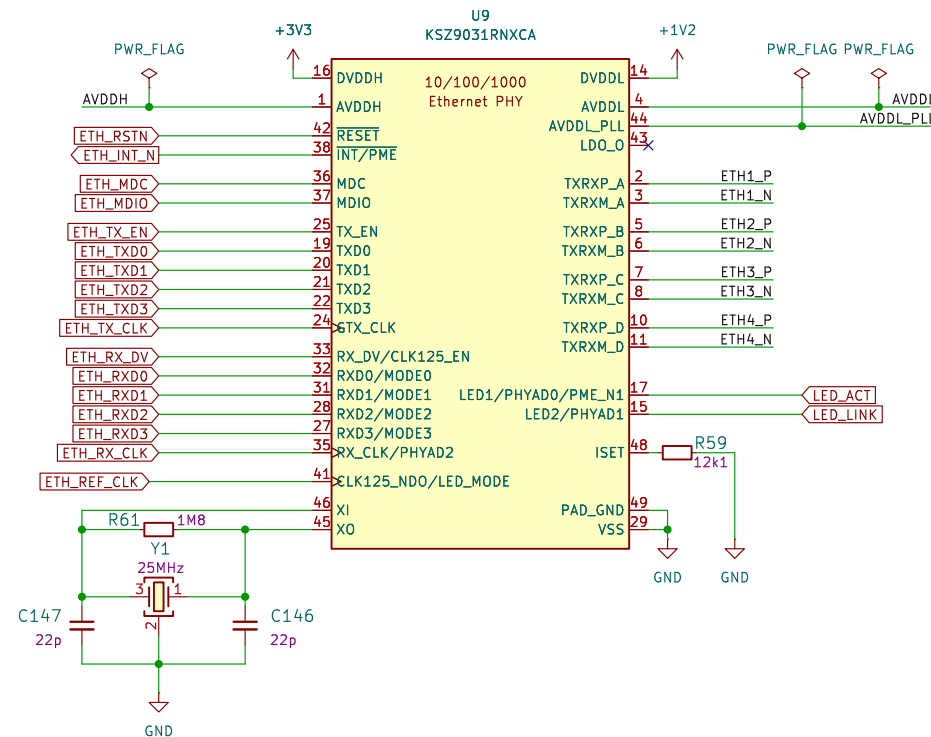
AVDDL_PLL decoupling



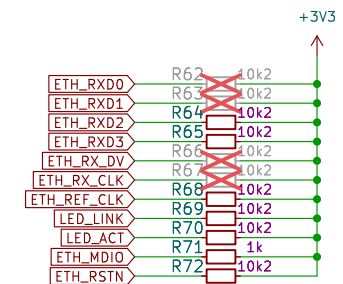
AVDDL decoupling



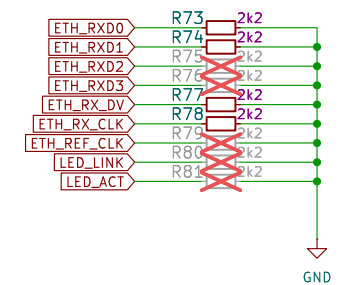
PHY



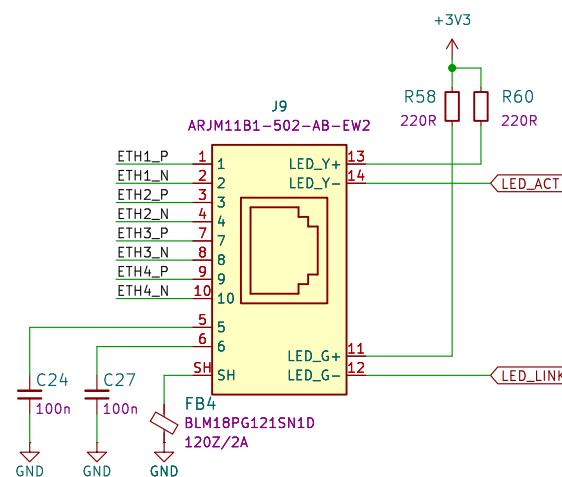
Pull up resistors

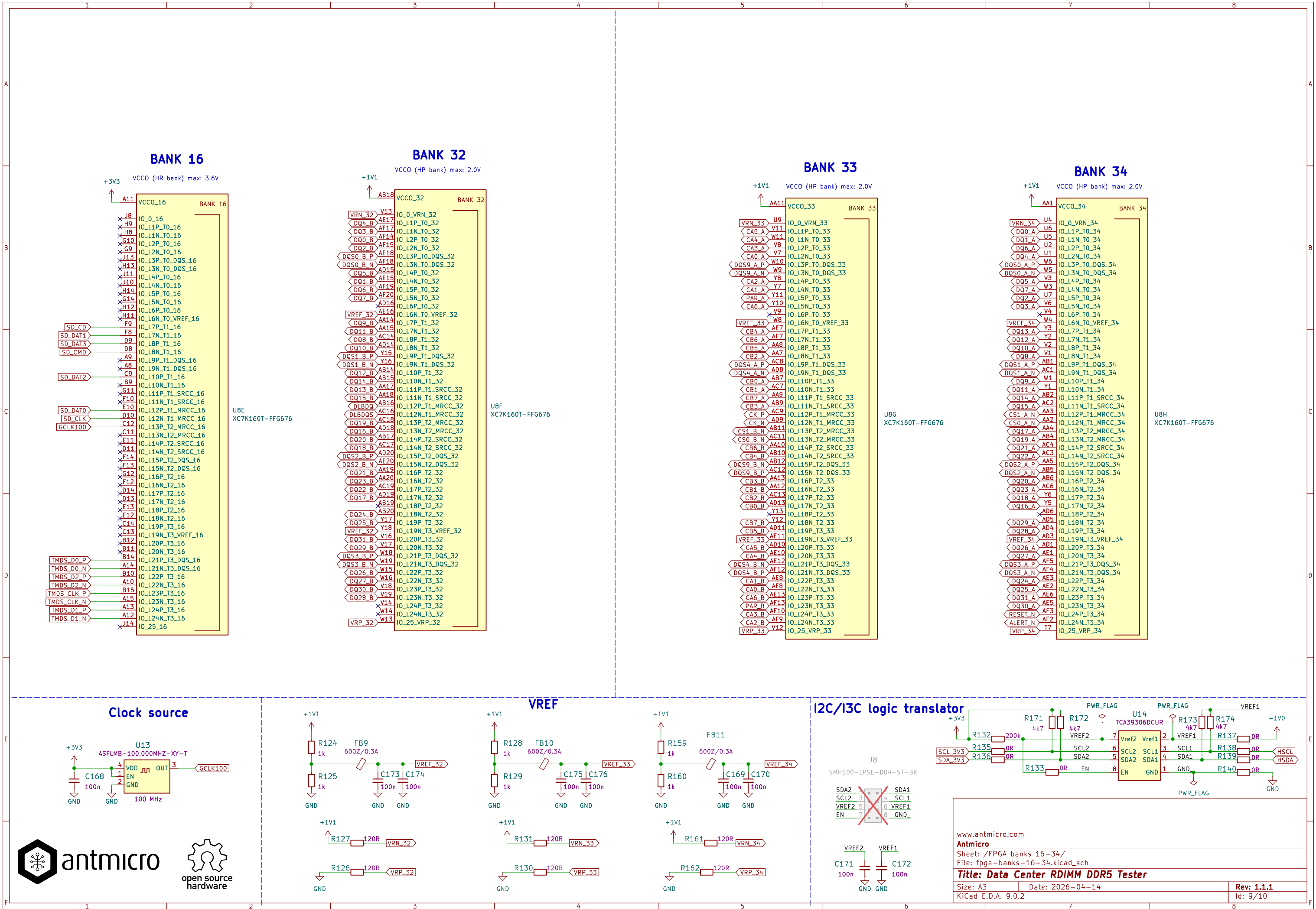


Pull down resistors

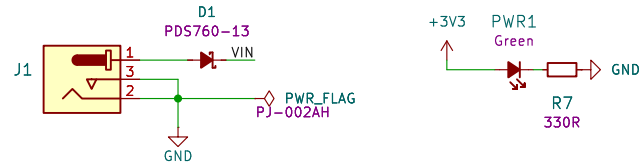


RJ45 Connector

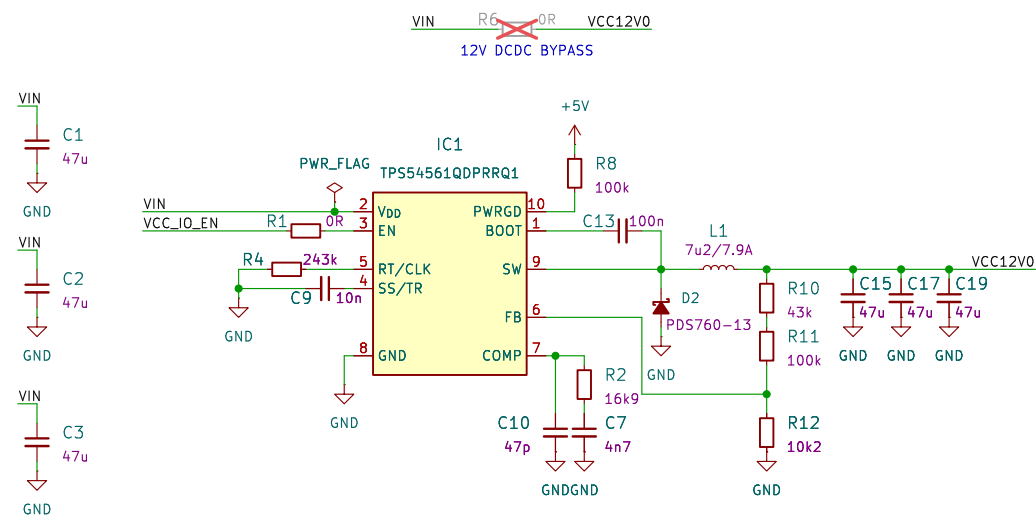




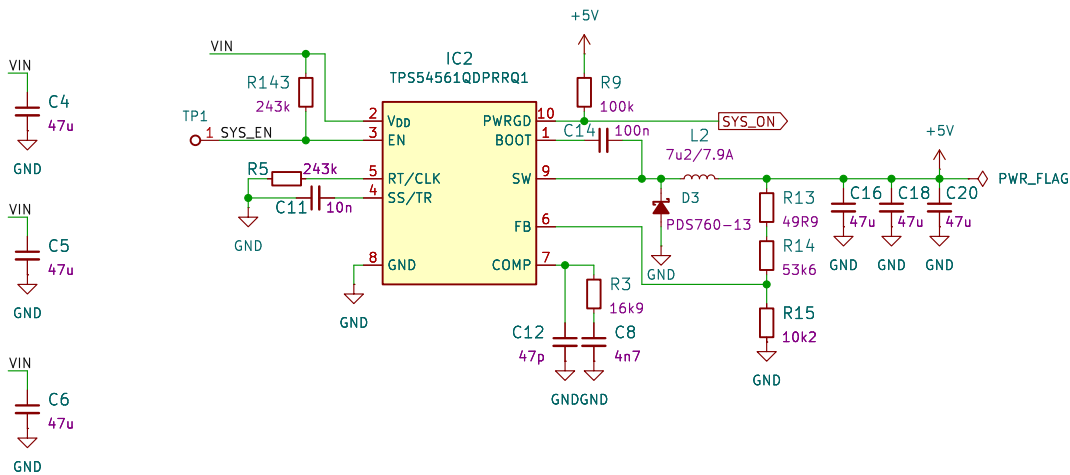
Input power connector



Main supply (12V 5A)



5V0 supply



VCCINT (1.0V 12A)

